

Module 10: OOP - Interfaces

Objective: To create and work with interfaces

System requirements: .Net 5.0

Tools: Visual Studio 2019

Duration: ~30min

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1 Creation of an interface

Based on the previous labs, an interface is to be created. This interface should give the implementing classes the ability to 'load' other vehicles.

Create an interface called `ILoadable` and add the following members to it:

- A property named `load`, in which any vehicle can be placed.
- A method (`Load(Vehicle)`), which is passed any vehicle and places the passed vehicle into the `load` variable
- A method (`Unload()`) which returns a vehicle and clears the `load` variable

2 Implementation

Let the `Ship` class implement the interface and write the method logic:

- - `Load()` shall test if the `load` property is already occupied and if it isn't, puts the passed vehicle into it (A vehicle is loaded).
- - `Unload()` should return the vehicle if the `load` is occupied and clear the cargo space. Set the `load` property to null for this.
- - Extend the `Info()` method of the `Ship` class with the information of a loaded vehicle.

3 Test

Extend the `Program` class by another method. This should take any two vehicles and test whether one of them is loadable. If this is the case, the other vehicle is to be loaded onto it.

Test this out considering different vehicle types.

```
None of the vehicles can load the other.  
Loading of 'Boeing' onto 'Titanic' successful.  
Cargo space of 'Titanic' is already occupied by 'Boeing'
```